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Letter

Long-term temporal variations of solar coronal rotation from 1976 to 2019

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ABSTRACT

Solar rotation is a pivotal driver of magnetic activities and dynamo processes, especially as the rotation rate varies with the atmospheric heights. To better understand the temporal variation of solar coronal rotation, the multi-frequency (245–15400 MHz) radio data from 1976 to 2019 covering solar cycles 21 to 24 are used. Based on the Higuchi fractal dimension and auto-correlation analysis techniques, the main results are as follows. (1) The regularity of solar mid-to-high frequencies (410–8800 MHz) is stronger, making these frequencies more suitable tracers for studying solar rotation variation; (2) Solar rotation inferred from radio emissions exhibits a significant frequency-dependent behavior, but the rotation rate derived at different frequencies shows a non-monotonic variation; (3) Solar cycles 23 and 24 exhibit a markedly larger jitter, which may be related to the evolutionary transition of the Modern Maximum and the accompanying changes in solar atmospheric structure. Our results provide an additional dynamical constraint on the evolution of the solar atmosphere during the decay phase of the Modern Maximum, complementing irradiance or flux-based evidence for long-term structural changes. This study offers new observational guidance for models linking photosphere, chromosphere and corona, and long-term solar cycle variability, with implications for heliospheric conditions and solar-stellar comparative studies.

1. Introduction

Solar rotation is one of the fundamental features, and it is the main driver of magnetic activities in the atmospheric layers [1–4]. The differential characteristics of the Sun, including the latitude-dependent differential rotation and radial variation with height, hold the key clues to understanding the interior dynamics and are related to coronal heating, solar wind acceleration, and space weather disturbances [5–8]. Since the 20th century, the academic community has gradually revealed the rotation laws from photosphere to corona through ground-based and space-based observations [9–20]. However, as the outermost atmosphere of the Sun, the corona's low plasma density and lack of stable tracking structures lead to many controversies regarding the observation of its rotational characteristics [21–24]. Long-term multi-frequency radio observational data provide a unique opportunity to investigate the rotational behavior across different heights, offering insights into how the corona couples to the underlying magnetic activity over multiple solar cycles [25–29].

Our understanding of solar rotation has been built upon several key methodologies. Early studies focused on the photosphere, primarily using the tracer method [30,31], which measures the rotation rate by tracking magnetic features across the solar disk. Newton and Nunn [32] pioneered the quantitative description of photospheric differential rotation by analyzing recurrent sunspots. This technique was later adapted for the corona by tracking features such as extreme-ultraviolet (EUV) coronal holes, which were found to exhibit nearly rigid rotation [33], and coronal bright points (CBPs), which helped refine rotation profiles [34–36]. Another key technique, the spectroscopic method [37–39], relies on measuring the Doppler shift of spectral lines. By analyzing the shifts in the H and K lines, Evershed [40] first hypothesized that the Sun's interior rotates faster than its surface. Modern instruments provide full-disk Doppler maps, confirming that rotation speed generally increases from photosphere to chromosphere [41]. To probe beneath the surface, helioseismology investigates the solar interior by observing the waves that propagate through it, revealing the rotation profile within the convection zone [42,43]. Finally, the flux modulation method [16,44–47] analyzes periodic variations in full-disk radiation as active regions rotate. This method is particularly powerful in the radio domain, as different frequencies probe different atmospheric heights [48,49].

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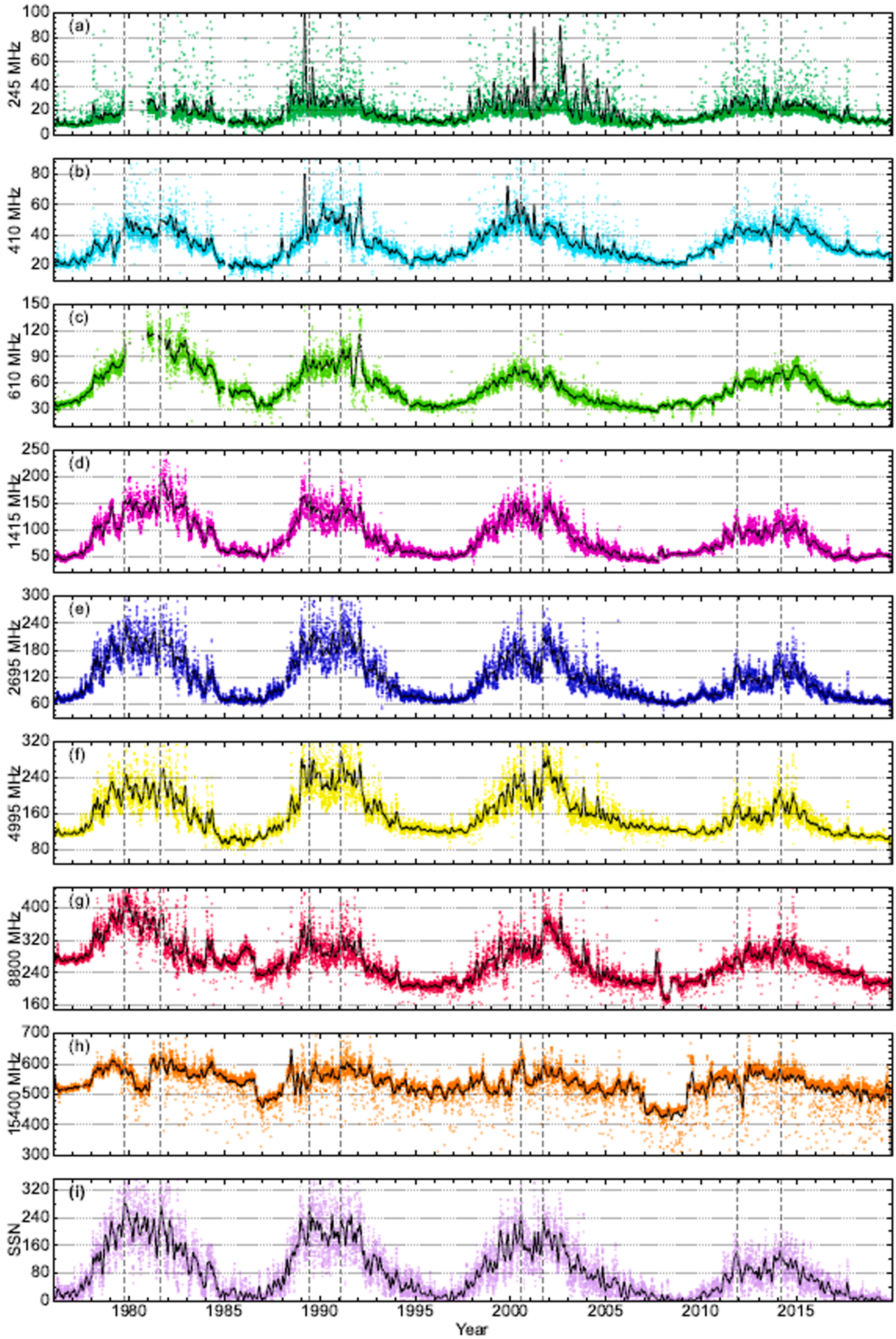

Fig. 1. The daily radio flux (colored dots) at eight frequencies from top to bottom: 245 MHz (green), 410 MHz (cyan), 610 MHz (bright green), 1415 MHz (pink), 2695 MHz (blue), 4995 MHz (yellow), 8800 MHz (red), and 15400 MHz (orange), along with the sunspot number (purple). The thin solid lines in the subgraphs are optimized smooth curves. The missing parts in the figure are null values in the original data.

These combined research efforts have established fundamental characteristics of solar rotation, such as its dependence on latitude and its modulation by the 11-year Schwabe cycle or 22-year Hale cycles [50,51]. Despite these advances, the radial gradient of coronal rotation, i.e., the rotation variation with height, remains a core issue of controversy, largely because stable tracking structures are scarce in the tenuous corona and different diagnostics sample different heights [52]. One group of studies suggests that the rotation rate increases with height (i.e., the rotation period becomes shorter). For example, Parker et al. [53] found that in the low-latitude region during solar cycle 20, between 1.1 and 1.5 solar radii ($R_{\odot}$), the coronal rotation rate increases with height; Vats et al. [25] used radio flux data (275–2800 MHz) to report that the outer corona rotates faster than the inner corona. Similarly, Sharma et al. [54], analyzing SDO/AIA data, found that the coronal rotation period significantly decreases with increasing temperature (a proxy of height). To characterize the differential rotation profile of the upper chromosphere using cleaned solar full-disc 17 GHz radio imaging from the Nobeyama Radioheliograph, Routh et al. [55] concluded that the upper chromosphere rotates faster than the photosphere at all latitudes. Conversely, a larger body of work indicates that the rotation rate decreases with height (i.e., the rotation period becomes longer). Lewis et al. [56] used SOHO/LASCO white-light observations and found nearly constant rotation in the height range of 2.5 to 15 $R_{\odot}$. Based on LASCO C1 Fe XIV emission, it was inferred that the outer corona is slightly slower than the inner corona; Mancuso and Giordano [39] further reported that, in the equatorial corona, the rotation period increases by about 0.2 days per solar radius. Critically, Bhatt et al. [28] reanalyzed the same radio dataset used by [25] with an improved fitting method and reached the opposite conclusion, questioning the robustness of earlier results. Most recently, Xiang et al. [29] reported that, despite complex temporal variations, the general tendency is a decreasing rotation rate with increasing height.

These disparate findings highlight the profound impact of diagnostic choice, analysis method, and data interval on the inferred height dependence of rotation. Therefore, a robust and uniform time-series approach is required to systematically extract rotation periods and quantify their temporal variations across multiple cycles. Auto-correlation analysis is a powerful method for identifying recurring patterns within a single dataset [57]. By measuring the correlation of a time series with a time-lagged version of itself, it effectively reveals dominant periodicities, making it well suited for determining synodic rotation periods from solar flux data [58,59]. In this study, we leverage a comprehensive multi-frequency (245–15400 MHz) solar radio flux dataset from the U.S. NOAA/RSTN network spanning four solar cycles (1976–2019) and apply a consistent auto-correlation framework to extract rotation periods across frequencies. This enables us to examine (i) whether the inferred sidereal rotation periods exhibit a systematic frequency (height) dependence, including possible non-linear behavior, and (ii) how their temporal variability evolves from cycle to cycle.

Finally, it is useful to place the studied interval (1976–2019) in the context of the roughly centennial Gleissberg cyclicity. Solar activity reached unusually high levels during the mid-20th century Modern Maximum (MM), after which the overall activity declined markedly from solar cycle (SC) 19 to 24, culminating in an exceptionally long and deep minimum between cycles 23 and 24 [60,61]. Importantly, the space-age observations largely coincide with the maximum and decay of the MM, implying that many commonly used activity proxies (including radio fluxes) have been recorded during a systematically weakening phase of solar magnetism. Therefore, part of the enhanced fluctuations and apparent anomalies in SC23–24 may reflect not only cycle-to-cycle variability, but also longer-term (Gleissberg-scale) changes in the solar atmosphere and magnetic-field topology during the decay of the MM.

2. Data and methods

2.1. Data

2.1.1. Data source

The solar radio flux data used in this study were obtained from the Sagamore Hill Solar Radio Observatory, which is under the National Oceanic and Atmospheric Administration (NOAA) of the United States, and the subsequent extended multi-station collaborative observation data from the Radio Solar Telescope Network (RSTN) of the U.S. Air Force. The data are archived in the public database of the National Centers for Environmental Information (NCEI).[1] As one of the core stations of the RSTN, this observatory has been continuously conducting solar radio observations since 1966. Its data are widely used in studies such as the solar activity cycle, coronal rotation, and radio emission mechanisms due to their long-term sequence integrity, multi-frequency coverage, and strict calibration procedures, providing continuous samples for verifying the correlation between radio emissions and the solar activity cycle.

In this study, eight radio frequencies were selected, namely 245 MHz, 410 MHz, 610 MHz, 1415 MHz, 2695 MHz, 4995 MHz, 8800 MHz, and 15400 MHz, covering the decimeter to centimeter wavebands and corresponding to different heights of the solar atmosphere. Although most of the radio data have strong continuity, there are some missing data. If the data missing rate is less than 10%, linear interpolation was used to complete the data before auto-correlation processing; for years with excessive missing data, linear interpolation will be used to fill in the gaps after auto-correlation processing. However, due to noise interference and the lack of rotational modulation in the 245 MHz and 15400 MHz frequencies, Singh et al. [59] suggested that the coronal region where the 245 MHz frequency band is located has an asymmetric structure, resulting in a lack of systematic rotational modulation characteristics in radiation changes. Bhatt et al. [62] reported significant noise interference in the 15400 MHz frequency band. Therefore, the data of these two frequencies were not used for rotation study.

The daily data of the International Sunspot Number (SSN) were obtained from the World Data Center for the Sunspot Index and Long-term Solar Observations (WDC-SILSO) of the Royal Observatory of Belgium,[2] with a time span from 1818 to the present [63–66]. As a core indicator for observing solar activity, the combination of the sunspot number and radio flux observations can more clearly illustrate the close relationship between radio data and solar activity.

2.1.2. Data preprocessing

Fig. 1 plots a comparison diagram of the daily radio flux at eight radio frequencies and the daily sunspot number, intuitively presenting the correlation of their temporal evolution. Different colors in this figure represent the daily radio data at different frequencies. The black thin lines are the optimized smoothing curves obtained using the optimized smoothing method proposed by [67]. The core of this optimized smoothing technique is to achieve the coordinated optimization of data fitting accuracy and curve smoothness by constructing an objective function, rather than the single local averaging used in traditional methods. The traditional smoothing method (13-month moving average) shows “reverse bias” and loses short-term key fluctuations. In contrast, the optimized smooth curve is highly consistent with the trend of the original

[1] https://www.ngdc.noaa.gov/stp/space-weather/solar-data/solar-features/solar-radio/noontime-flux/sagamore-hill/.

[2] https://www.sidc.be/SILSO/datafiles.

data while eliminating high-frequency noise. This advantage is also reflected in Fig. 1 of this work.

After optimized smoothing, the short-term fluctuations of the radio flux (such as small fluctuations during the maximum period of a certain solar cycle) are completely retained. A gray dashed line is marked at the smoothed peak of the sunspot number in the figure to compare with the peak of the radio data. It is observed that the peaks of these radio frequencies are almost completely in phase with the peaks of the sunspot number, while the traditional moving average would obscure such details. In this study, optimized smoothing effectively filters out the daily-scale noise in the radio data caused by instrument interference and interplanetary scintillation, making the long-term evolution trends (such as the 11 yr Schwabe cycle) of the eight radio frequency curves and the sunspot number curve in Fig. 1 more prominent and reducing the interference of noise.

As can be seen from Fig. 1, the temporal evolution of radio fluxes at different frequencies and the international SSN shows remarkable synchrony. Specifically, both the radio fluxes and the SSN exhibit periodic fluctuations following solar cycles 21–24. For example, during the maximum of solar cycle 22 (around 1989), cycle 23 (around 2001), and cycle 24 (around 2014), the radio fluxes at all frequencies and the SSN reach their peaks simultaneously; during the minimum periods (such as 1976, 1986, 1996, and 2008), both of them drop to their troughs. This confirms that radio fluxes can serve as tracers of chromospheric, transition region and coronal activity indices, responding to the changes of solar magnetic structures dominated by the photosphere.

Among the numerous indicators for studying solar activities, radio fluxes display a unique and crucial advantage. When the number of sunspots decreases significantly and approaches zero, the radio flux does not drop to an extremely low level. On the contrary, it maintains a relatively low value that is far from zero. This characteristic is of great significance, as it means that even during the minimum of the solar cycle, the radio flux can still more accurately reflect the state of solar magnetic field. Compared with the situation where the sunspot number almost loses its indicative function during the activity minimum, this performance of radio flux makes it an important and reliable observational indicator for studying solar activities, especially during the low-activity periods, providing a valuable and effective basis for scientists to continuously and in-depth understanding the laws and changes of solar activities.

The radio fluxes at frequencies of 410, 610, 1415, 2695, 4995, and 8800 MHz have a stronger correlation with the SSN, and their curves highly coincide with the SSN curve in shape. The low-frequency 245 MHz radio flux is more discrete and has more significant curve noise due to the influence of interplanetary scintillation and radio interference. The 15400 MHz radio flux slightly deviates from the main trend of the sunspot number due to local activity interference.

2.1.3. Fractal analysis

Fractal dimension is a metric for measuring the irregularity and randomness of time series [62,68–73]. In the context of solar activity, fractal analysis has been applied to global solar indices and solar radio fluxes to characterize their scale-dependent temporal variability [62,74,75]. For a time series composed of independent and identically distributed random variables, D approaches 2, whereas for a smooth periodic signal, D approaches 1. Therefore, a larger D indicates a stronger random component, while a smaller D indicates a more regular or periodic temporal variation. Following previous solar radio-flux studies, $D = 1.0$–1.4, $D = 1.4$–1.7, and $D = 1.7$–2.0 are interpreted here as low, moderate, and high randomness, respectively [62].

In this study, we used the Higuchi method to calculate the fractal dimensions of the radio flux time series at eight frequencies, namely 245 MHz, 410 MHz, 610 MHz, 1415 MHz, 2695 MHz, 4995 MHz, 8800 MHz, and 15400 MHz. The Higuchi method estimates the fractal dimension of a discrete time series directly in the time domain by examining the scale dependence of the curve length [68]. This method has been successfully applied in previous studies of solar radio flux variability and coronal rotation [44,62,69].

For a discrete time series $X(1), X(2), \ldots, X(N)$, a new time series is constructed for each scale k and initial point m as

$$X_m^{(k)} = \left\{ X(m), X(m+k), X(m+2k), \ldots, X\left(m + \left\lfloor \frac{N-m}{k} \right\rfloor k\right) \right\}, \tag{1}$$

where $m = 1, 2, \ldots, k$, and $\lfloor \cdot \rfloor$ denotes the floor function. The curve length corresponding to $X_m^{(k)}$ is calculated as

$$L_m(k) = \frac{N-1}{\left\lfloor \frac{N-m}{k} \right\rfloor k^2} \sum_{i=1}^{\left\lfloor \frac{N-m}{k} \right\rfloor} |X(m+ik) - X(m+(i-1)k)|. \tag{2}$$

The mean curve length at scale k is then given by

$$L(k) = \frac{1}{k} \sum_{m=1}^{k} L_m(k). \tag{3}$$

If $L(k)$ follows a power-law relation with k, namely $L(k) \propto k^{-D}$, the Higuchi fractal dimension D can be estimated from the slope of the linear fit between $\ln L(k)$ and $\ln(1/k)$.

The calculated fractal dimensions of each frequency band were displayed in a two-dimensional distribution according to time and frequency, and a fractal dimension contour map was drawn in Fig. 2. It should be noted that, in order to better display the fractal dimension variations across different frequencies, the x-coordinates between each frequency were uniformly distributed instead of being aligned according to the original frequency magnitude. This ensures that the temporal evolution trends of fractal dimensions for different frequencies can be clearly distinguished, and avoids information congestion in the high-frequency segment (caused by excessively large frequency intervals) or information compression in the low-frequency range. In Fig. 2, the ordinate represents the time span, covering 1976–2019; the abscissa represents the radio frequency; and the contour lines represent the changes in the fractal dimension values. Blue corresponds to a lower fractal dimension, indicating that the radio flux is more regular and has lower randomness. The redder the color, the larger the fractal dimension, indicating higher randomness of the radio flux time series. During the minimum years of the solar activity cycle, there is a certain increase in randomness, which further indicates that the radio flux is significantly modulated by the solar activity cycle. Near the maximum years of the solar activity cycle, there is no significant change in randomness between low-frequency and high-frequencies. However, for 410 MHz, 610 MHz, 1415 MHz, 2695 MHz, 4995 MHz, and 8800 MHz, the degree of modulation by the solar activity cycle is greater than that of the radio flux at high (15400 MHz) and low (245 MHz) frequencies. Near the maximum years, the fractal dimensions of these frequencies significantly decrease, and the randomness weakens.

The 2695 MHz radio flux is close to the 10.7-cm radio emission (2800 MHz). The 10.7-cm radio emission has relatively low irregularity and is suitable as an observational object for the study of coronal rotation [76–79]. For the 245 MHz and 15400 MHz frequencies, the distribution of their fractal dimension contour lines is highly irregular, with disordered color changes and no obvious regular patterns. This means that the radio flux time series of these two frequencies has extremely high randomness, and the fractal dimensions at these two frequencies do not show a clear and consistent correlation with the solar cycle, lacking the clear temporal variation trend observed in the medium-frequencies. Thus, it indicates that these two frequencies have limitations in reflecting the overall laws of solar activity, and the use of data from these two frequencies may introduce large errors and interference in studies that require capturing stable modulation signals.

2.2. Auto-correlation analysis

In this study, the auto-correlation analysis is employed to extract the solar rotation period from the solar radio flux. Correlation analysis

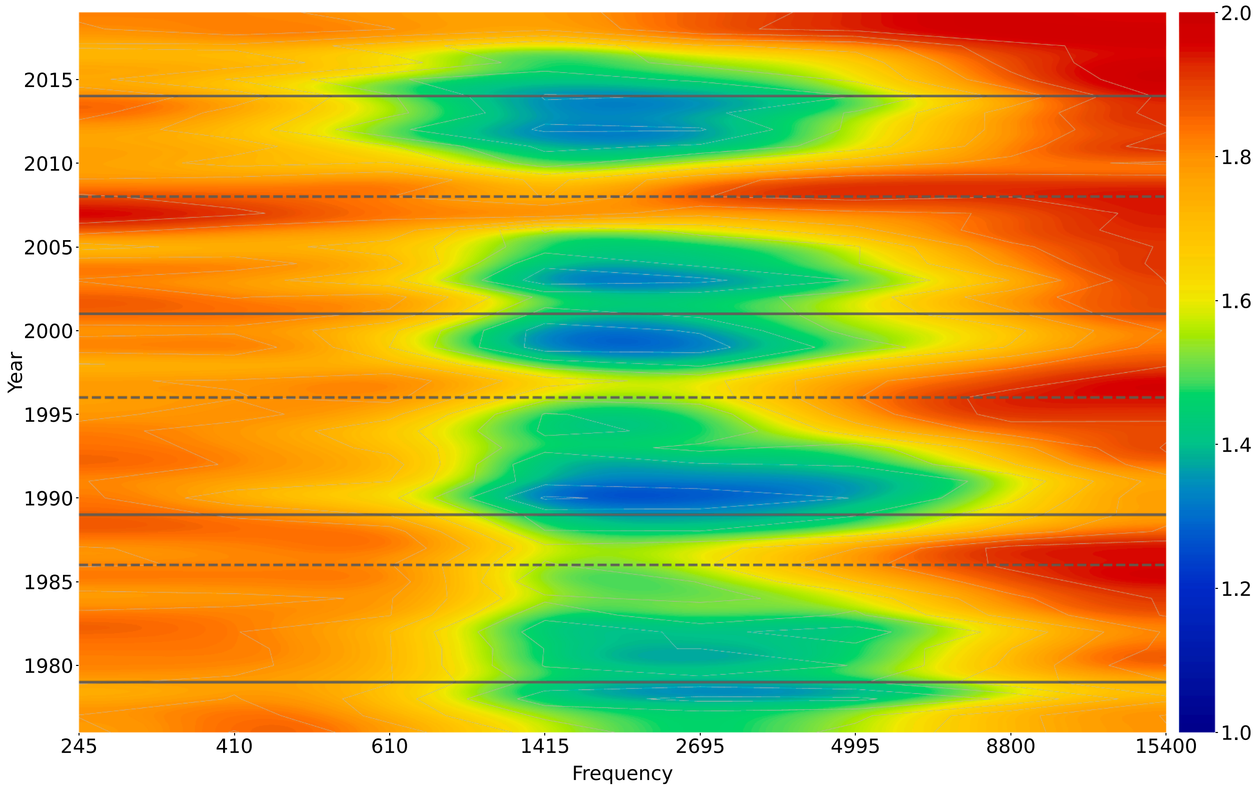

Fig. 2. Contour map of the fractal dimension of radio flux at 245 MHz, 410 MHz, 610 MHz, 1415 MHz, 2695 MHz, 4995 MHz, 8800 MHz, and 15400 MHz as a function of year (covering four solar activity cycles from 1976 to 2019). The solid and dashed horizontal lines represent the solar maximum periods and solar minimum periods, respectively.

generally refers to the certain correlation existing between two sets of variables, and the degree of correlation determines the similarity between the two sets of signals. The correlation coefficient ranges from [−1, 1]. When the two sets of signals are completely the same, the correlation coefficient is 1, indicating a positive correlation between the two sets of variables; when the correlation coefficient is -1, it means a negative correlation between the two sets of variables. In this paper, the auto-correlation procedure is used to study the periodic characteristics of a set of signals themselves. The auto-correlation analysis, as a powerful tool, has been widely applied in solar activity fields to uncover the periodic features, thereby enabling in-depth investigation of important phenomena such as solar rotation. Numerous scholars have utilized the auto-correlation techniques to extensively explore the rotation characteristics of different atmospheric layers of the Sun. The auto-correlation of a time series refers to the delay correlation of the sequence at different time lags. For a given solar radio flux time series $(x_0, x_1, x_2, \dots, x_{n-1})$, there can be $(n - l)$ pairs of observations (where n is the length of the sequence and l is the lag interval): (x_0, x_l), (x_1, x_{l+1}), (x_2, x_{2+l}), …, (x_{n-l-1}, x_{n-1}). If the first observation in each pair is regarded as the first variable and the second observation in each pair is regarded as the second variable, the autocorrelation coefficient $P_x(l)$ as a function of the lag l can be calculated by Eq. (4). In Eq. (4), $\bar{x}$ is the mean value of the sequence, that is, $\bar{x} = \frac{1}{n}\sum_{k=0}^{n-1} x_k$.

$$P_x(l) = P_x(-l) = \frac{\sum_{i=0}^{n-l-1} \left(x_i - \bar{x}\right)\left(x_{i+l} - \bar{x}\right)}{\sum_{i=0}^{n-l} \left(x_i - \bar{x}\right)^2} \tag{4}$$

The auto-correlation analysis requires careful attention to data preprocessing. When a sequence contains significant errors, the autocorrelation coefficient tends to approach zero; if short-term correlations are present, the coefficient tends to be close to one; and after the lag exceeds a critical value, the coefficient generally converges to zero. For the nonstationary sequences, the overall trend may cause the autocorrelation coefficient not to decay as expected, necessitating the removal of trend components beforehand [80].

In this study, the synodic rotation period was determined by applying a Gaussian fit to the first secondary maximum of the auto-correlation function. To measure the confidence level of the results, a confidence line was added as a standard. To improve accuracy, points near the peak of the first secondary maximum were selected for fitting, and the fitting effect was evaluated using R^2; if R^2 is negative or 0, it indicates that the fitting failed; the closer R^2 is to 1, the better the fitting effect. In the image obtained after using the auto-correlation analysis, the auto-correlation results are significantly affected by the trend term. As shown in Fig. 3, the auto-correlation results after removing the trend are more significant. For the vast majority of the analyzed years, the synodic rotation period was successfully derived directly from the first secondary maximum. Only in a very few exceptional years, where the first secondary peak fell below the 95% confidence level, did we select subsequent significant peaks (e.g., the second, third, or even later peaks) for the Gaussian fit. This approach ensures a robust synodic rotation period determination with a higher confidence level [27].

3. Analysis results

3.1. Yearly sidereal rotation periods

This study applies the multi-frequency solar radio flux data from 1976 to 2019 to extract the synodic rotation period through the auto-correlation analysis. To analyze the Sun's intrinsic rotation period, we converted the observed synodic periods to sidereal periods using Eq. (5) [51,80–82].

$$T_{\text{sidereal}} = \frac{365.26 \times T_{\text{synodic}}}{365.26 + T_{\text{synodic}}}. \tag{5}$$

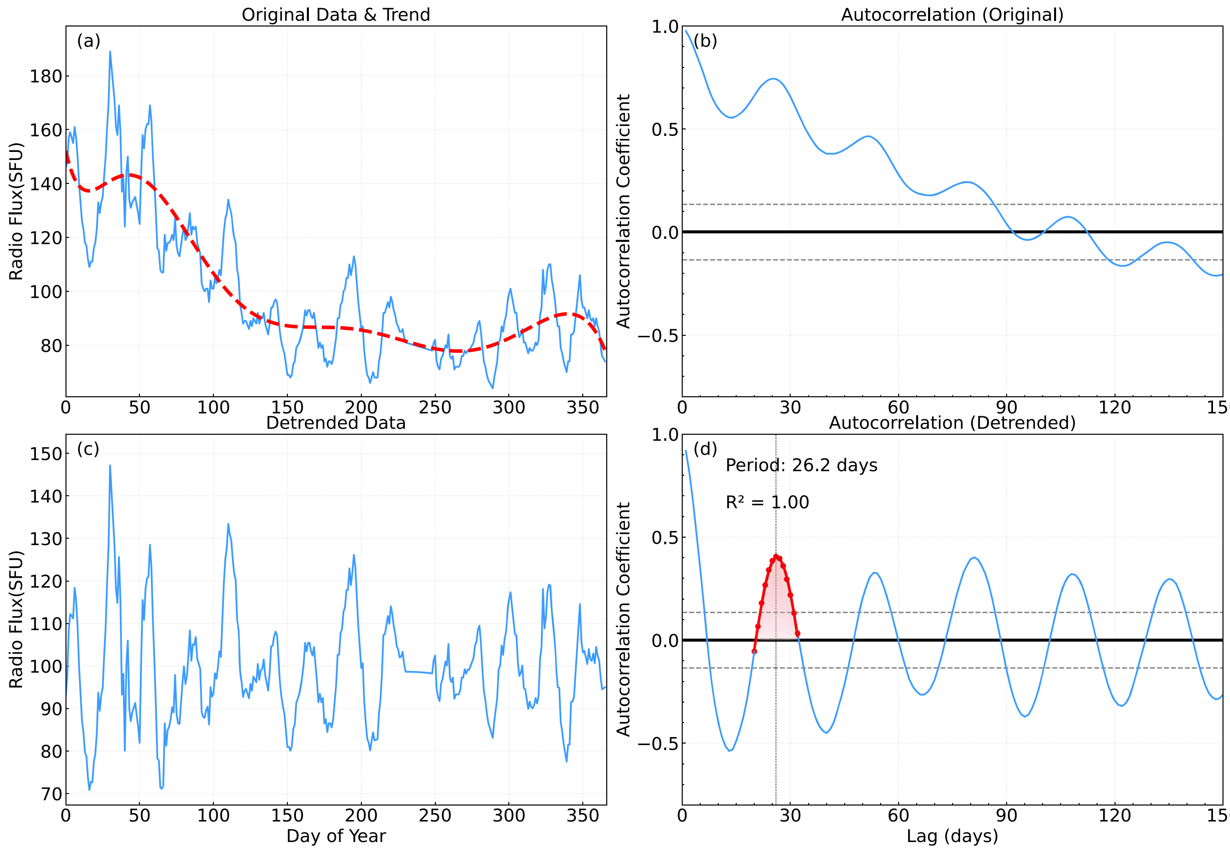

Fig. 3. Panel (a) shows the original radio flux data at 1415 MHz in 1992, with the dashed line representing the linearly fitted trend term. Panel (b) shows the results of the auto-correlation coefficient calculated from the original data, displaying a time lag of 0–150 days. Panel (c) shows the image of the original data after removing the trend. Panel (d) shows the auto-correlation curve of the detrended data. The red points are the points selected for Gaussian fitting, and the red curve is the auto-correlation coefficient after Gaussian fitting. The gray dashed lines in Panels (b) and (d) represent the 95% confidence level.

Herein, $T_{synodic}$ denotes the synodic rotation period, and $T_{sidereal}$ denotes the sidereal rotation period. The converted sidereal rotation period data are presented in Table 1, which covers annual data across six valid frequencies: 410 MHz, 610 MHz, 1415 MHz, 2695 MHz, 4995 MHz and 8800 MHz.

During the data processing, it was found that some years had data missing or quality issues: the 610 MHz observation data in 1980 were insufficient due to observation interruptions, failing to meet the minimum sample size requirement for the auto-correlation calculation; the data for years such as 1987, 2007, 2008, 2009, 2010 and 2018 are affected by factors like instrument noise and atmospheric interference, leading to severe fluctuations in the solar radio flux data. It was impossible to fit the auto-correlation curves with a confidence level higher than 95% for some frequencies. To ensure the continuity of the time series, for years with a data gap of less than 10%, we used linear interpolation to fill the series before the auto-correlation analysis. For years with larger gaps, we performed the interpolation on the resulting period data using values from adjacent years.

From Table 1, the solar rotation period inferred from radio emissions exhibits a significant dependence on observing frequency. The sidereal rotation rate derived from the radio emission observed at 2695 MHz has always been at a leading level, with its period values being the smallest among all frequencies in the same year. For instance, the period at 2695 MHz was only 22.61 days in 1998 and 22.52 days in 2015, which are significantly shorter than those at other frequencies; the adjacent 4995 MHz frequency also maintained a relatively fast rotation, with an average annual period of 25.22 days. The rotation rates of the low-frequencies (410 MHz and 610 MHz) were relatively slower. Among them, the average annual period was 25.73 days for 410 MHz, 25.43 days for 610 MHz, showing a trend that the rotation slightly accelerates as the frequency increases in the low-frequencies. In contrast, the average annual period of the high-frequency 8800 MHz band reached 25.12 days, with the largest fluctuation amplitude (standard deviation of 2.05 days). For example, the period at this frequency was as high as 29.43 days in 2005 and dropped to 23.22 days in 2011, with an inter-annual difference of nearly 6.21 days, reflecting that the rotation of the high-frequency band is more significantly disturbed by the magnetic activities in the chromosphere.

In different years, the sidereal rotation periods at various frequencies are not constant but follow a pattern of variation with years. In some years, the rotation periods at all frequencies are generally smaller, indicating an overall acceleration of sidereal rotation; in other years, the periods are generally larger, meaning the rotation slows down relatively. This variation may be related to the height of the sources corresponding to different frequencies or the phases of solar cycles.

3.2. *Temporal evolution across frequencies*

From the perspective of temporal evolution from 1976 to 2019, the sidereal rotation periods derived from radio emissions at all analyzed frequencies show regular fluctuations synchronized with the solar activity cycles (Fig. 4). During the solar maximum years, the rotation periods of most frequencies tend to be longer. Taking 2695 MHz as an example, its period was 27.04 days in 1989, longer than the 25.23 days in the 1986 minimum period. In contrast, during the minimum periods, the periods of all frequencies are generally at relatively low levels. For the low frequencies, the rotation period shows a decreasing trend with increasing frequency. As can be seen from Fig. 4 (a) and

Table 1
Sidereal rotation period estimated from multi-frequency radio flux (1976–2019). The quantities are given in days.

Year	410 MHz	610 MHz	1415 MHz	2695 MHz	4995 MHz	8800 MHz
1976	27.81	26.87	26.01	24.45	24.79	23.62
1977	24.19	24.45	23.49	25.40	25.40	22.83
1978	25.14	25.40	24.71	24.79	24.88	25.15
1979	–	25.57	26.27	25.92	25.83	24.66
1980	26.95	–	25.31	25.05	25.14	25.01
1981	25.14	25.57	27.12	24.62	24.71	24.92
1982	–	24.36	24.71	24.62	24.62	24.40
1983	23.84	25.23	24.71	25.40	23.49	23.27
1984	24.79	–	25.75	25.31	27.55	27.69
1985	24.71	24.19	24.45	24.27	24.10	25.06
1986	24.88	26.78	25.14	25.23	–	–
1987	28.23	–	23.49	23.66	–	23.09
1988	29.59	–	27.30	26.61	26.78	–
1989	26.61	27.64	27.55	27.04	26.95	27.39
1990	25.05	24.45	24.88	24.88	25.05	24.66
1991	23.92	23.40	24.19	23.75	23.49	23.88
1992	–	–	24.45	25.14	24.71	24.80
1993	26.01	25.40	24.27	25.92	26.44	27.26
1994	25.31	24.71	25.92	26.44	26.09	27.77
1995	25.92	25.31	24.27	24.36	25.05	24.40
1996	26.27	26.35	25.31	26.27	24.79	–
1997	25.31	25.14	24.53	24.19	23.18	20.70
1998	27.72	25.66	23.84	22.61	21.55	24.99
1999	25.31	25.14	25.57	25.40	26.01	22.56
2000	27.21	24.27	23.75	23.40	27.98	28.67
2001	–	24.88	25.14	24.97	25.31	23.88
2002	26.27	23.92	24.71	25.57	26.35	24.58
2003	–	25.23	25.31	25.49	26.78	28.80
2004	–	24.79	25.75	26.01	26.09	26.75
2005	26.05	25.05	24.97	24.79	25.40	29.43
2006	24.53	25.14	23.75	23.31	23.84	22.56
2007	–	28.57	28.74	23.40	–	24.84
2008	27.47	27.04	25.92	25.83	–	–
2009	24.01	24.45	24.71	–	25.49	–
2010	26.01	30.18	–	30.10	–	–
2011	27.04	24.19	21.90	22.17	22.26	23.22
2012	24.19	24.79	24.88	24.88	24.71	25.88
2013	27.47	27.12	26.01	25.83	26.09	27.77
2014	24.36	24.19	24.27	24.19	24.62	25.18
2015	24.88	24.62	24.19	22.52	24.62	25.19
2016	22.78	26.35	27.72	26.18	27.89	28.03
2017	24.27	24.62	25.83	25.31	25.49	24.14
2018	26.52	–	30.01	–	–	–
2019	26.52	25.31	25.40	23.49	24.79	22.38
Mean	25.73	25.43	25.26	24.97	25.22	25.12
Std.Dev	1.45	1.34	1.45	1.38	1.38	2.05

(b), the rotation speed at 410 MHz is slower than that at 610 MHz, and the rotation speed at 610 MHz is slower than that at 1415 MHz. These three frequency bands all originate from the corona, which indicates that within this height range, the higher the altitude, the smaller the rotation speed. However, this pattern does not hold in the subsequent subfigures. Among the interwoven curves of adjacent frequencies, the variation amplitude of the sidereal rotation period with time for high frequencies seems to be smaller. To obtain more accurate results, we need to analyze the relationship between the rotation periods of these frequencies.

3.3. Variation of the rotation period with time and frequency

Solar rotation inferred from radio emissions exhibits a significant dependence on observing frequency. Fig. 5 shows the temporal and spatial distribution of rotation periods in different frequency bands (data processed with 3 yr smoothing method), where the dot-dashed line and dashed line represent the solar minimum and maximum years, respectively. The figure clearly shows that the rotation period at high frequencies is much shorter than that at low frequencies on the whole. It is worth noting that within each solar cycle, the rotation speeds of the high and low frequency bands are not synchronized, but show an anti-correlated "seesaw" trend: when the rotation inferred from the low-frequency radio emissions is slower, that inferred from the high-frequency emissions may be relatively faster, and vice versa.

More specifically, on a short-time scale, the rotation period may increase or decrease with increasing frequency. Similar patterns were also observed in the analysis of [59]: from 1994 to 1996, the rotation period slightly increased with increasing frequency; while from 1997 to 1999, it decreased with increasing frequency. Karachik et al. [83] proposed that coronal rotation rates may vary with altitude within the same corona coordinate system. Vats et al. [25] suggested that rotation periods decrease with increasing altitude, while [28] argued for increased rotation periods at higher altitudes. These studies all indicated the linear increases or decreases in rotational rates. However, our analysis reveals a different pattern: maximum rotation rates occur at specific altitudes, with rates decreasing as altitude decreases or increases.

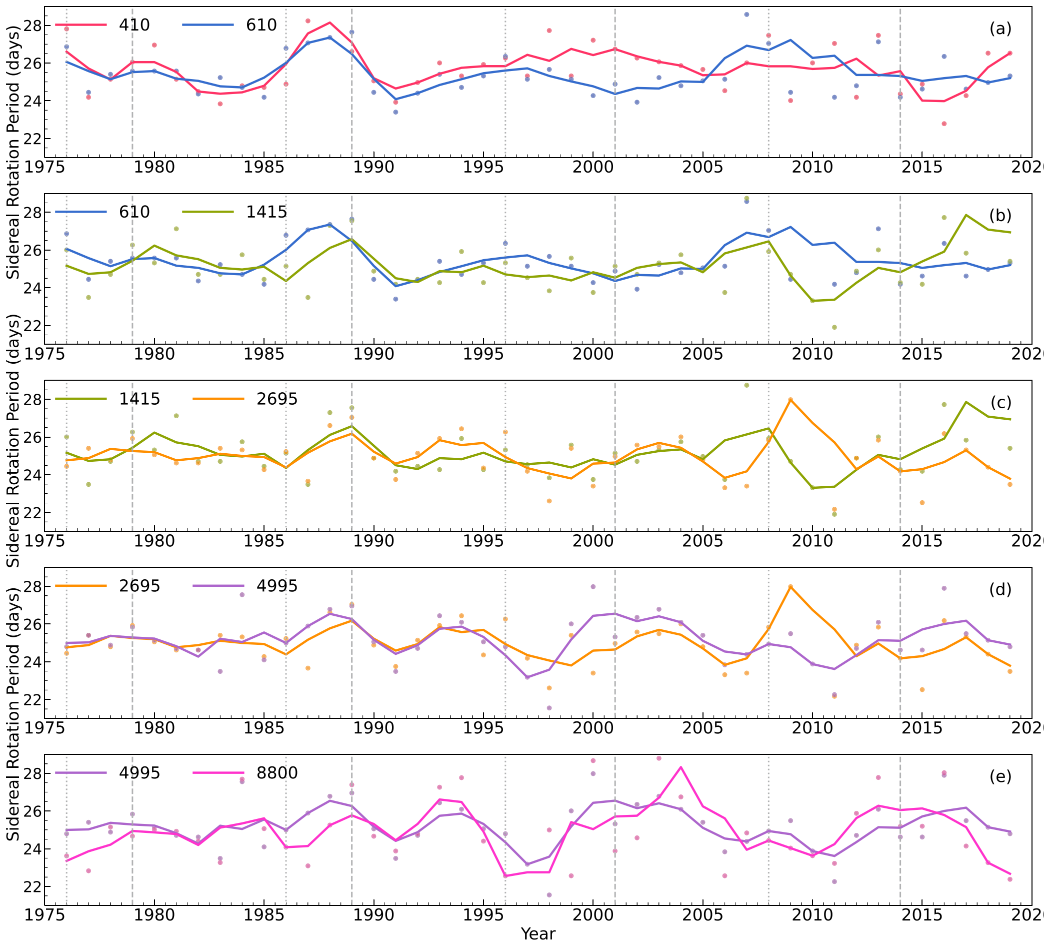

Fig. 4. The variation trends of sidereal rotation periods with time derived from radio emissions observed at 410, 610, 1415, 2695, 4995, and 8800 MHz annually from 1976 to 2019. The solid lines in the plot represent data smoothed over a 3 yr period. To facilitate easier comparison of sidereal rotation periods at different frequencies, the data from adjacent frequencies are combined and displayed, arranged from low to high frequencies from top to bottom. The gray dash-dot line represents the solar maximum periods, while the dashed line indicates the solar minimum periods.

3.4. *Rotation trend during different solar cycles*

To reveal the variation trend of rotation period in different solar cycles, we analyzed solar cycles 21–24 separately. We used the average value of each frequency as scatter points to fit the curve of rotation period versus frequency. As shown in Fig. 6, several solar cycles exhibit non-monotonic variations in the rotation period as a function of frequency. Among them, solar cycles 21 and 24 show that the rotation speed accelerates with increasing frequency, which is consistent with the results of [29]. However, solar cycle 23 presents a more pronounced non-linear trend with a larger amplitude. This may stem from a combination of unique factors: the polar magnetic field during the minimum of solar cycle 23 was much weaker than in other solar cycles [39,84–86], and given that coronal rotation is highly sensitive to differences in polar magnetic field strength and hemispheric magnetic flux emergence rates [87], this weakness likely plays a key role. Its exceptionally long period (12.3 years) and unusually intense global magnetic activity further contribute to significant temporal variations in the coronal rotation period of cycle 23 (such as sudden accelerations or decelerations), ultimately allowing different physical mechanisms to be fully manifested at various altitudes and amplifying the differences in rotational rates between high and low frequencies.

The angular velocity data from different heights (including photosphere, chromosphere, and corona layers), as presented in Fig. 3 of [55], clearly show non-monotonic increases in rotational rates across various altitudes. Another interesting observation from the figure is that when the rotational gradient of the inner corona increases (i.e., the rotation period accelerates more rapidly with height), the rotational gradient of the outer corona decreases (i.e., the rotation period changes more gradually with height), and vice versa. Mancuso and Giordano [39] also reported this phenomenon, with their study indicating an anti-correlation in the radial gradient between the inner and outer corona: the inner corona is dominated by magnetic pressure with tightly connected plasma, while the outer corona is characterized by open magnetic field lines. Badalyan and Obridko [88] found that the equatorial rotation rate increases during the minimum between even and odd solar cycles, while it decreases during the minimum between odd and even cycles. Additionally, the absolute value of the differential rotation gradient was observed to increase during even cycles. These factors may contribute to the enhanced amplitude of odd cycles compared to their preceding even cycles, potentially explaining the Gnevyshev-Ohl rule, where odd cycles are generally more active than the preceding even cycles.

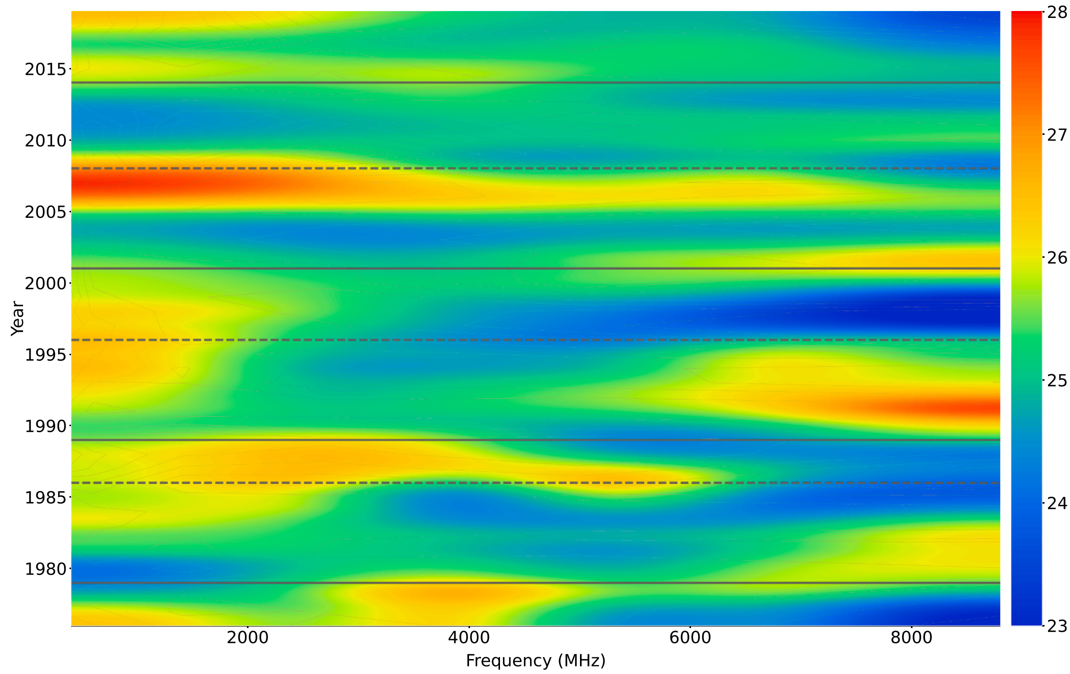

Fig. 5. The distribution map of the sidereal rotation period inferred from radio emissions versus observing frequency at six frequencies ranging from 410 to 8800 MHz during the study time interval from 1976 to 2019. The data in the figure underwent a 3 yr smoothing method. The solid and dashed horizontal lines represent the solar maximum years and solar minimum years, respectively.

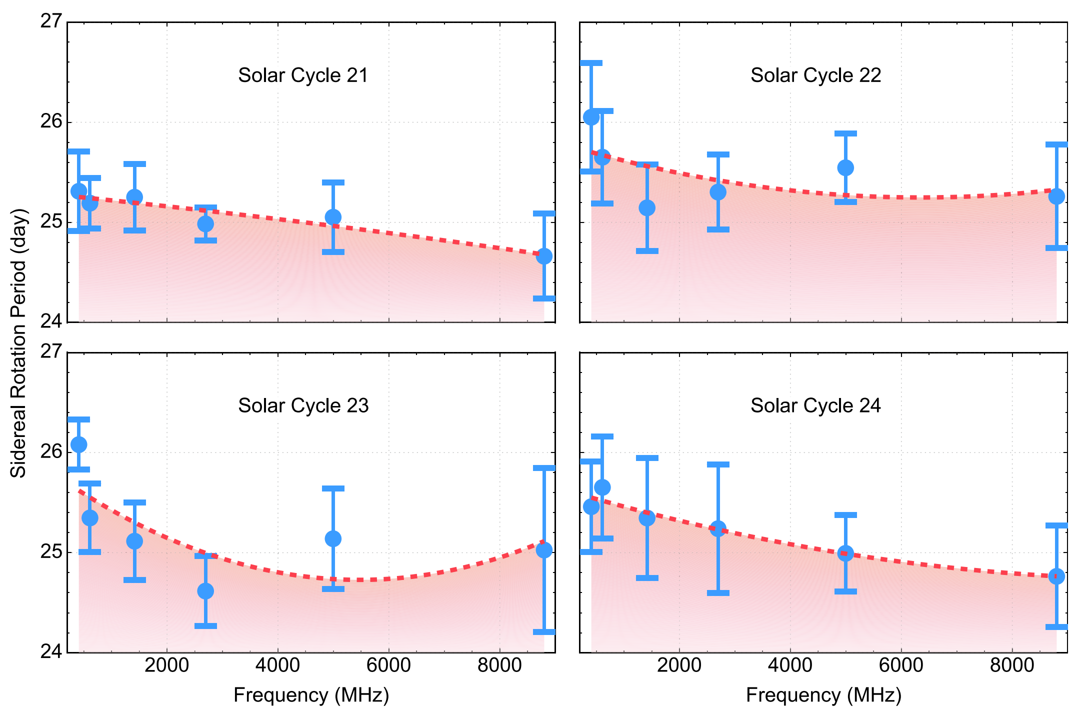

Fig. 6. The temporal variation of sidereal rotation period with frequency during solar cycles 21–24. Blue data points represent the average rotation periods at different frequencies, with error bars indicating standard errors. The red dashed line shows the quadratic polynomial fitting curve.

4. Discussions and conclusion

4.1. Discussions

The multi-frequency radio data provide a unique tomographic view of the Sun's atmospheric rotation, revealing both persistent, long-term structures and significant short-term dynamics. The foundation of our interpretation is the principle that radio emission frequency maps to atmospheric height. It is the magnetic field that ultimately controls the solar cycle and generates solar activity [41]. From the perspective of radiation characteristics, as the wavelength increases from centimeter to decimeter and further to meter waves, the source region of radio emissions progressively shifts to higher altitudes [48,49,89]. Although the precise heights of these emissions are challenging to determine, it is well established that the low-frequency radio emissions arise from higher layers of the solar atmosphere, while high-frequency emissions originate from lower layers.

The faster rotation observed in higher layers like the chromosphere and corona is thought to be intrinsically linked to the ubiquitous magnetic field, which is rooted in subsurface layers and plays a critical role in maintaining phenomena like the meridional flow and the near-surface shear layer [90]. For instance, magnetic features such as sunspots are known to rotate at different speeds than the surrounding plasma, with variations suggesting different anchoring depths within the convection zone [91]. The faster rotation at higher surface layers, the chromosphere and corona, was also supposed to be related to the ubiquitous magnetic field rooted in the subsurface layers [92,93].

The regularity of solar radio flux, quantified by fractal dimension, is a persistent characteristic tied to the physical conditions of each atmospheric height. Our fractal analysis reveals that the lowest (245 MHz) and highest-frequency (15400 MHz) bands exhibit the most irregular behaviors (fractal dimension > 1.6), consistent with findings by [62]. This is physically intuitive; the low-frequency signals are disturbed by plasma turbulence in the upper corona, while the high-frequency signals are affected by localized, transient photospheric events. In contrast, the mid-frequency bands become significantly more regular during solar maximum, suggesting that strong and large-scale magnetic fields create more coherent rotating structures, yielding a cleaner signal. However, it is worth noting that although low frequencies and high frequencies exhibit strong irregularity, they are also modulated by solar magnetic field [60,94–96].

When averaged over the entire 44 yr study period, the solar rotation profile inferred from the multi-frequency radio emissions exhibits a non-linear dependence on frequency/height: the rotation period does not change monotonically with atmospheric altitude, but shows a broad mid-frequency extremum, with the fastest rotation typically appearing around the mid-frequency bands (e.g., 2695–4995 MHz) and comparatively slower rotation toward both lower and higher frequencies. This large-scale behaviour can be understood as the combined outcome of multiple, competing processes acting across atmospheric heights. For example, at the upper boundary, the outer corona can be decelerated by the torque applied by the outflowing solar wind on open magnetic field lines, which may contribute to longer periods inferred from the highest-altitude tracers. This interpretation remains consistent with long-term studies that reported a net tendency for coronal rotation to slow with increasing height, while also emphasising that the frequency/height profile is not strictly linear across the sampled layers [28,29]. Although the multi-decadal mean profile is non-monotonic, the rotation profile on shorter timescales is highly dynamic. We propose this variability is driven by the dynamic magnetic coupling between the corona and the fast-rotating near-surface shear layer, mediated by the evolving magnetic fields of active regions. This coupling is not constant; its strength varies, causing the radial gradient of rotation to reverse periodically. Our analysis shows this transition occurs on a timescale of approximately 3–5 years, where periods of rotation speeding up with height are followed by periods where it slows down. This timescale intriguingly aligns with the solar quasi-biennial oscillation (QBO) [97–99], suggesting a potential physical link that warrants future investigation.

The mechanism of variable coupling manifested in an exceptionally strong and persistent manner during solar cycle 23. Instead of merely reversing the monotonic gradient, the coupling was so intense that it produced a distinct non-monotonic peak in rotation speed in the mid-frequency band (2695–4995 MHz). For extended periods, this mid-atmospheric layer rotated significantly faster than the layers below it, a dramatic departure from the long-term trend. The unique conditions of solar cycle 23, notably its exceptionally weak polar magnetic field and significant post-reversal changes in heliospheric dynamic pressure [84,100], likely amplified the efficiency of angular momentum transfer from the shear layer. Thus, solar cycle 23 serves as a powerful illustration of how short-term magnetic conditions can temporarily override the general long-term rotational structure, leading to profound anomalies [39].

The pronounced jitter observed in SC23–24 can be further interpreted in the framework of the centennial Gleissberg cyclicity and the decay of the Modern Maximum (MM). During the MM, solar activity reached unprecedented levels, but then declined markedly from cycle 19 to cycle 24, culminating in an exceptionally long and deep minimum between cycles 23 and 24. This multidecadal weakening phase is not merely a rescaling of activity amplitude; rather, it is associated with structural changes in the solar atmosphere and magnetic topology [60,101–103]. In particular, multi-frequency flux observations indicate that longer-wavelength fluxes increased relative to shorter wavelengths from the 1960s to the 2010s, which has been interpreted as an effective change in the altitude distribution of plasma density and a shrinking of the corona as solar activity weakened during the decay of the MM [104].

Such long-term atmospheric restructuring provides a natural pathway to amplify cycle-dependent and frequency-dependent differences in radio-based rotation tracers. Moreover, the long-term relation between sunspots and upper-atmospheric radiative output is not constant over the MM: recent results suggest that sunspot activity dominates relative to EUV irradiance during the growth and maximum of the MM, whereas EUV irradiance dominates relative to sunspots during low-activity cycles and during the decay of the MM. Since EUV irradiance mainly originates from chromospheric and low-coronal plages/faculae, this implies a Gleissberg-scale modulation of the spot–facula balance [61,105,106].

Because different radio frequencies sample different atmospheric heights, a systematic change in the spot–facula/plage balance and coronal structure during the late MM (SC23–24) can plausibly enhance the interannual fluctuations and the apparent anomalies of rotation periods inferred from multi-frequency radio fluxes, in addition to cycle-specific factors such as the unusually weak polar field during cycle 23. Accordingly, the markedly enhanced jitter inferred in SC23–24 can be interpreted as a dynamical manifestation of the late-phase MM evolution: as the atmosphere restructures and the relative role of plage/faculae-related emission increases, the rotation tracers sampled by different frequencies become more sensitive to year-to-year changes in the dominant emitting structures. In this sense, the SC23–24 jitter is not only cycle-to-cycle variation, but may reflect a Gleissberg-scale transition of the solar atmosphere during the decay of the MM [60,61].

It is important to note the data limitations of this study. Due to high noise levels and weak rotational modulation signals in the 245 MHz and 15400 MHz frequencies [59,62], these two frequencies were excluded from the rotation analysis, potentially limiting insights into rotation in the outermost corona and the transition region. Additionally, for years with a data missing rate of <10%, linear interpolation was used to complete the dataset. Although a 3 yr smoothing was applied to reduce bias, this approach may still introduce minor errors.

4.2. *Conclusion*

Based on the long-term multi-frequency solar radio observations from 1976 to 2019 covering solar cycles 21–24, combined with fractal dimension analysis and auto-correlation analysis, we draw the following main conclusions:

1. **Irregularity is frequency-dependent.** The highest (15400 MHz) and lowest (245 MHz) frequency bands show the strongest irregularity, whereas the mid-frequency range (410–8800 MHz) becomes significantly more regular (especially during solar maximum), making these frequencies more suitable tracers for studying rotation variations.
2. **The overall rotation profile is non-linear across frequency/height.** Solar rotation inferred from different frequencies shows a pronounced frequency-dependent behaviour, and the derived rotation period does not vary monotonically with fre-

quency/height; instead, the fastest rotation tends to occur at specific mid-frequency bands, with the rotation becoming slower toward both lower and higher frequencies.

3. **SC23–24 exhibit markedly enhanced jitter, likely linked to the MM decay.** The larger inter-annual fluctuations in SC23–24 may be associated with the evolutionary transition of the Modern Maximum toward a weaker Sun and accompanying structural changes in the solar atmosphere and magnetic topology, in addition to cycle-specific factors such as the unusually weak polar field during SC23.

Data availability

The solar radio flux data used in this study are available from the NCEI at https://www.ngdc.noaa.gov/stp/space-weather/solar-data/solar-features/solar-radio/noontime-flux/sagamore-hill/.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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